

EcoOrbit: methane and pollutant intelligence

CH4 case studies, quality-screened atmospheric columns and supporting land-use classification

Scope. Methane case targets: Stanford Arizona controlled-release location (2022 verified study), Hassi R Mel Algeria approximate field center, and Jeddah Wadi Al-Asla landfill. New case-specific CH4 scenes are pending. The 24 observed US industrial points and 24 school controls support an additional land-use classifier.

Methane case evidence

Case	Coordinate	Evidence status
Stanford Arizona	32.82182, -111.78577	2022 published releases; suggested 2024-25 period unverified
Hassi R Mel	about 32.9, 3.2	Published regional methane study; not a surveyed stack
Jeddah landfill	21.64419, 39.39026	Landfill location documented; three proposed 2024 dates unverified

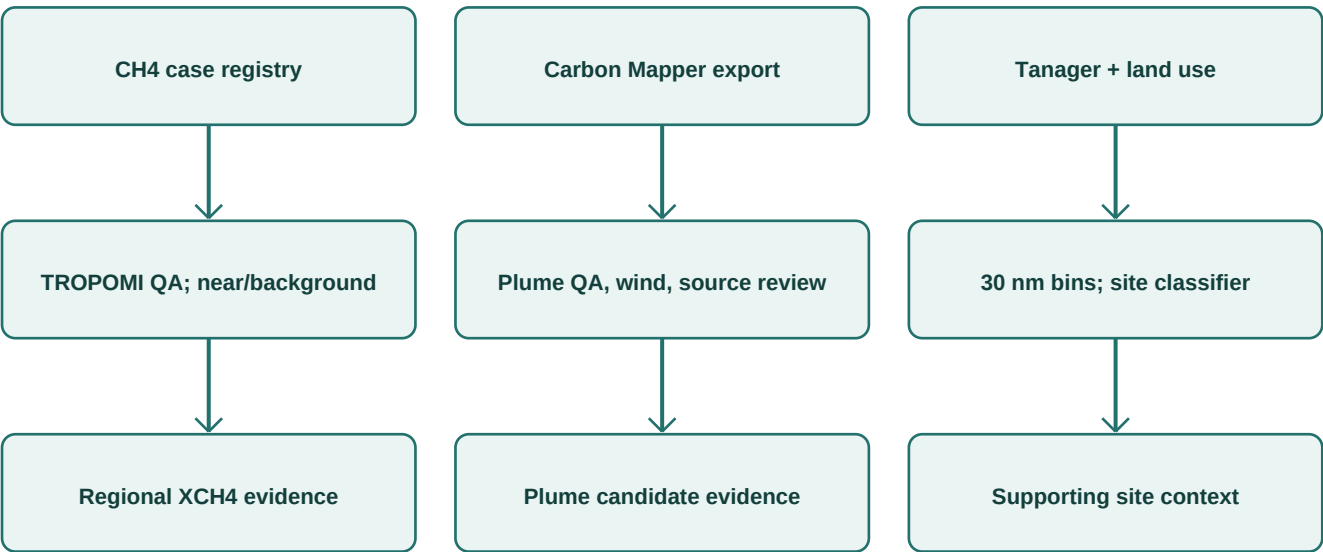
TROPOMI CH4 workflow: XCH4 ppb, QA > 0.5; compare 0-25 km to 25-80 km ring, requiring three valid pixels each. A contrast is regional evidence, never source attribution or an emission rate. Optional Carbon Mapper CH4 and CO2 plume exports add quality, wind, instantaneous gas-specific kg/h and uncertainty. No target CO2 plume is bundled.

Real published plume examples

Location / gas	Date	Provider-reported rate
Brazil landfill / CH4	29 Sep 2024	Preliminary plume: 2,836 kg/h
India power generation / CO2	1 Nov 2024	Preliminary source: 838,000 kg/h
EMIT example / CH4	20 Apr 2024	3,610.58 +/- 377.95 kg/h

These are source-linked published examples, not emissions at the three proposed case sites. The first two image IDs are given in provider image filenames; the original article omits exact origin coordinates and uncertainty. See [data/published_plumes.csv](#).

Pipeline



Preprocessing and calibration

Input	Applied processing	Interpretation
Planet Tanager L2A	Provider radiometric, atmospheric and geometric corrections; HDF no-data/cloud/cirrus masks; 30 nm reflectance bins and three spectral indices.	Site spectral features, not a raw sensor calibration.
Landsat 8/9 L2	QA_PIXEL mask; Celsius = $DN \times 0.00341802 + 149 - 273.15$. 180 m site median minus 600-1200 m annulus.	Land surface contrast; thermal information coarser than 30 m output grid.
Sentinel-2 L2A	SCL clear pixels; two-date NDVI= $(B08-B04)/(B08+B04)$; NDBI= $(B11-B08)/(B11+B08)$.	Land change can reflect season and cover.
Sentinel-5P L2	NO2 QA >0.75; SO2 QA >0.5; nearest valid pixel center within 15 km.	Atmospheric column mol/m2, not stack emission or ground concentration.

Supporting land use and observed coverage

The 60 site features (57 reflectance bins plus NDVI, NDBI and MNDWI) are processed by imputation, scaling, eight-component PCA and balanced logistic regression. A fixed stratified split trains on 36 sites and tests on 12, two per class in each city. Model selection uses training-only cross-validation; schools are narrow comparison controls.

Evaluation	Sites	Accuracy	ROC AUC
All-city test	12	0.917	0.944
Train-only CV	36	0.722	-
Earlier city holdout	48	0.750	0.804

Primary test confusion matrix (true school/industrial by predicted school/industrial): $\begin{bmatrix} 5 & 1 \\ 0 & 6 \end{bmatrix}$. The 424 additional registry points have no extracted spectra and are excluded from training. Schools are a narrow built-environment negative class. A new script retrieves OSM green-space and commercial polygon candidates; expanded test metrics are pending source imagery QA.

Coverage of added observations

Result	Detroit	Jacksonville	Rochester
Landsat valid sites / site-date rows	3 / 6	8 / 16	8 / 16
Two-date Sentinel-2 paired sites	2	6	0
NO2 and SO2 in supplied file	8 sites each	Download pending	No image-week items

The available Landsat median site-minus-nearby-land contrasts are +0.66 C (Detroit), +1.45 C (Jacksonville) and +4.00 C (Rochester), on different dates and backgrounds. They must not be ranked as factory heat-emission rates.

Jacksonville Tanager green reference: 12 separated, clear vegetation-rich image patches. Cluster-wide median NDVI 0.770 versus median 0.305 across eight industrial-site neighborhoods (difference +0.465). The cluster median is not 12 independent patch measurements; no verified park geometry or emission inference is claimed.

Limits and ESG reporting status

A TROPOMI cell spans kilometres and can include multiple factories and other sources. The nearest pixel center is a geographic lookup only. Negative SO2 retrievals near background are measurement noise, not negative physical emissions. ERA5 10 m wind and CAMS modeled air provide regional context, not a factory plume or discharge rate.

GRI 305-1/305-2 and the GHG Protocol require organizational boundaries, fuel/process and electricity records, factors and uncertainty for Scope 1/2 tCO2e. GRI 305-7 requires measured or justified NOx/SOx mass estimates. Satellite columns and wind provide context, not these quantities. The generated report is an evidence-gap demonstration, not a conforming disclosure.

Carbon accounting and interactive results

A documented facility activity CSV drives Scope 1 and Scope 2 tonnes CO2e per declared period; the dashboard shows unavailable when records are missing. Carbon Mapper CO2 or CH4 plume kg/h is instantaneous and cannot be annualized without persistence and a reporting boundary. Interactive tabs filter gas, case, date, NO2, SO2, particulate model context and aerosol products.

Reproducibility

Open the included dashboard with: `streamlit run dashboard.py`. The notebook and scripts reproduce the workflow. Source snapshots, exact scene IDs, QA thresholds, dates and missing statuses accompany the CSV outputs. See README.md and docs/methodology.md for the complete sources and correction steps.

Sources: Planet Tanager docs; USGS Landsat Collection 2 Surface Temperature; EPA FRS API; Copernicus Sentinel-5P NO2 and SO2 product readmes; Microsoft Planetary Computer; Open-Meteo CAMS and ERA5 access. Full URLs are in docs/methodology.md.